TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**AI-powered NDT@3Dprinters**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials		**Knowledge Services**	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics / Automation Machinery	Mechanical Automation Technologies	R&D / Engineering Services	Product Quality Management Technologies
Concept of Proposed Technology	The proposed technology is an integrated, AI-powered non-destructive testing (NDT) solution for industrial 3D printers. It combines multiple NDT methods with advanced data fusion and joint inversion to provide automated, reliable, and traceable quality assurance during and after additive manufacturing processes. The system is designed to be embedded into existing 3D printing equipment and to support sustainable, resource-efficient production. The technology can be applied to metal, ceramic, and polymer 3D printing in sectors such as aerospace, automotive, energy, medical devices, and high-performance tooling. Typical uses include in-line quality assurance for safety-critical parts, certification support for new materials and processes, process development and optimization, and implementation of circular-economy strategies by extending component life and reducing waste.			

Rationale for Joint R&D Project	The proposed R&D project is needed because current industrial 3D printing lacks a reliable, integrated, and automated quality assurance solution (1), (2). Different 3D-printing processes (metal, ceramic, polymer) require tailored NDT outcomes (3), but no commercially available system combines multi-modal NDT, data fusion, AI-based interpretation, and sustainability assessment (4). As a result, defect rates are high, certification is difficult, and a significant amount of material, energy, and machine time is wasted (5), which limits broader market adoption and conflicts with circular-economy and climate goals. Collaboration is required because the necessary expertise, infrastructure, and use-cases are distributed internationally and cannot be built up in Korea alone within the targeted timeframe. Fraunhofer IKTS brings advanced NDT, AI, and system integration know-how, while Korean partners provide cutting-edge additive manufacturing processes, industrial testbeds, and access to the Korean and Asian markets. This international cooperation is therefore essential both to accelerate development and to address global challenges such as resource efficiency and climate impact in additive manufacturing.
Key Words	1. AI-powered non-destructive testing (NDT) 2. Data fusion for defect detection 3. In-line quality assurance for 3D printing 4. AI-based defect classification 5. Sustainable additive manufacturing monitoring

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	Development of an integrated set of technologies covering sensors, data processing, AI, and system integration for NDT in industrial 3D printing: 1. Multi-modal NDT sensor modules: - Application-specific sensor setups for thermography, infrared transmission/reflection, acoustic emission, X-ray backscattering, and ultrasound. - Robust data acquisition hardware suitable for harsh industrial 3D-printing environments (temperature, vibration, dust). 2. Data fusion and joint inversion algorithms - Physics-based models and algorithms to jointly invert multiple NDT signals into consistent 2D/3D defect and material maps. - Uncertainty quantification and Probability of Detection (POD) tools for reliable quality statements. 3. AI-based defect analysis and quality assessment software - Machine-learning models for automated defect detection, classification, and quality grading across different materials and printer types. - Software platform for visualization, reporting, and integration of NDT results into digital quality records. 4. Printer integration and real-time process feedback - Hardware and software interfaces between the NDT system and industrial 3D printers. - Control algorithms that adjust process parameters in (near) real time based on NDT feedback to reduce scrap and optimize quality. 5. Sustainability and life-cycle assessment (LCA) tools - Methods and software modules to quantify material, energy, and waste reduction enabled by the NDT system. - Decision-support tools to select process settings that balance part quality, cost, and environmental impact.			
R&D Target Specifications	Target performance specifications for the project are: - Detect internal defects in metal and ceramic AM parts with an equivalent diameter of ≥ 0.5 mm (wall thickness up to 30 mm) with POD 90/95 (90% probability of detection at 95% confidence). - Detect internal defects in polymer AM parts with an equivalent diameter of ≥ 1.0 mm with POD 90/95. - Defect size estimation accuracy within ±20% of true size. - False positive rate (defect indicated but not present) < 5%. - System availability (uptime) in industrial operation ≥ 95%. - In-line inspection that adds ≤ 10% overhead to the total build time for a typical industrial			

print job.

- Fully integrated operation with at least 3 different 3D-printing technologies (e.g. metal LPBF, ceramic AM, polymer AM) on industrial testbeds.
- 80% reduction of failure parts compared to the current state of the art in 3D-printing quality assurance.
- At least 50% reduction in scrap material and 30% reduction in energy consumption per accepted part, enabled by early defect detection and process feedback.
- Increase technology readiness from TRL 4 to TRL 7.
- Delivery of a validated prototype system (hardware + AI-based software) ready for pilot deployment in Korean industry.

R&D Plan

The project follows an application-driven approach: (1) jointly define requirements and architecture with Korean partners, (2) develop and integrate multi-modal NDT hardware and AI-based analysis tools, (3) demonstrate and validate the system in industrial 3D-printing environments in Korea, including POD and LCA. Work progresses from lab validation (TRL 4–5) to industrial pilot demonstrations (TRL 6–7).

Phase	Period (months)	Key tasks & scope	Main methodologies	Expected milestones
1	0–6	Pre-study / joint specification	Requirements engineering, market & standards review, benchmarking, concept design	M3: Joint requirements & use-cases agreed; M6: System architecture frozen
2.1	7–18	Core technology development (sensors, data fusion, AI)	Experimental NDT, multi-physics modelling, ML model development, SW engineering	M12: Lab prototypes of NDT modules; M18: Integrated lab prototype (TRL 5)
2.2	19–30	System integration with printers and Korean testbeds; initial industrial pilots	Hardware/software integration, hardware-in-the-loop testing, design of experiments on process parameters	M24: First pilot at Korean site; M30: Stable pilot system (TRL 6)
2.3	31–36	Validation, optimization, POD & LCA; preparation for commercialization in Korea	POD studies, LCA, reliability testing, standards mapping, guideline development	M33: Completed POD & LCA reports; M36: Demonstrated TRL 7 and final toolkit

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer IKTS will act as technical research institute, responsible for the scientific development of the AI-powered NDT system for industrial 3D printing. Fraunhofer IKTS will mainly be responsible for: - Multi-modal NDT development - Data fusion and joint inversion - AI-based analysis and software platform - System integration concepts - Performance validation and sustainability assessment Specific tasks to be performed by Fraunhofer IKTS: - Lead the pre-study and requirements engineering together with Korean partners (use-case definition, inspection targets, performance requirements). - Develop and validate lab-scale prototypes of NDT sensors and data acquisition systems. - Implement and refine data fusion, inversion, and AI models using lab and pilot data. - Support the mechanical and electrical integration of NDT modules into Korean industrial 3D printers. - Conduct joint pilot tests and analyse results to optimize the inspection concept and process feedback loops. - Prepare technical documentation, guidelines, and training material to enable technology transfer to Korean industry. Resources to be provided: - Access to Fraunhofer IKTS laboratories, NDT equipment, and test infrastructure for additive manufacturing. - Dedicated research staff (NDT experts, AI/data scientists, software engineers, systems engineers, and project management). - Existing software and modelling tools for NDT simulation, signal processing, AI development, POD, and LCA. - Support for IP management, exploitation planning, and preparation of joint publications and dissemination.
Expected R&R of Korean SME Partner	The Korean SME is expected to act as the industrial lead in Korea and the primary commercialization partner. The Korean SME will mainly be responsible for: - Providing industrial 3D-printing processes and products (metal, ceramic and/or polymer) as target applications. - Leading the engineering of the industrial prototype, including mechanical design and mounting of NDT modules on the printers. - Developing and implementing interfaces between the NDT system and the SME's printer controllers, and factory IT systems. - Adapting and packaging the technology into a marketable product/service for Korean

	customers (e.g. UI localization, service concept, maintenance procedures). - Provide pilot production lines and real components for testing, including baseline data on current scrap and quality levels. - Conduct field trials under typical industrial operating conditions. - Participate in performance validation, including reliability tests, and customer acceptance tests. The Korean SME should provide: - Access to 3D-printing equipment, auxiliary systems, and production facilities. - Dedicated engineering and operational staff (process engineers, quality engineers, operators, IT staff) to support integration and testing. - Materials, test parts, and production time for experimental campaigns and validation runs. - Internal investment and management support for scaling and commercialization in Korea. - Act as the lead Korean organization in the joint R&D project and coordinate with other Korean partners (e.g. universities, additional SMEs). - Participate in joint planning and review meetings, sharing requirements, data, and feedback with Fraunhofer IKTS.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	1. Technological achievements
	- Delivery of an integrated AI-powered, multi-modal NDT system for industrial 3D printers, demonstrated at TRL 7 in Korea.
	- Verified 80% reduction of failure parts compared to the current state of the art, through reliable in-line quality assurance.
	- Demonstrated sustainability benefits via LCA: reduced scrap, lower energy use per accepted part, and extended component lifetime.
	
	2. Commercialization potential
	- Transfer of the technology to Korean SMEs and industry, enabling them to:
	  o Offer NDT@3D printing solutions and quality-assured AM services.
	  o Differentiate through high-reliability, certifiable 3D-printed parts in safety-critical markets (aerospace, mobility, energy, medical).
	- Creation of a product and service portfolio (inspection systems, software licenses, maintenance, and data-driven services) with clear routes to domestic and later global markets.
	- Strengthening of Korea's position in the global AM value chain, including potential export of NDT-enhanced 3D-printing solutions to Asia and beyond.
	
	3. Broader industrial and economic impact
	- Support for circular-economy strategies: less material waste, fewer rejected parts, and longer component life through better quality control.
	- Productivity gains and cost reductions in Korean manufacturing by cutting rework, scrap, and certification effort.
	- Job creation in Korea in areas such as AM production, quality engineering, NDT services, software integration, and maintenance.
	- Contribution to climate and resource-efficiency goals by lowering the environmental footprint of 3D-printed products.
	- Establishment of internationally aligned practices and know-how in Korea, improving readiness for global standards, certifications, and cross-border production networks.

6. References

(1) The Application, Challenge, and Developing Trends of Non-destructive ...

(2) In-process monitoring and non-destructive evaluation for metal additive ...

(3) Review on additive manufacturing and non-destructive testing

(4) A Systematic Review of Advanced Sensor Technologies for Non-Destructive ...

(5) Scalable in situ non-destructive evaluation of additively manufactured ...

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization